PART A : STRUCTURED QUESTIONS [65 MARKS]

INSTRUCTION : ANSWER ALL FOUR (4) QUESTIONS.
Please write your answer in the answer booklet.

QUESTION 1 (10 MARKS)

Given the following scenario,

"A recreational centre near the lake in Universiti Teknologi Malaysia (UTM), provides many facilities for customers to use. One of the facilities is bicycle rental service. Customers can rent a bicycle to ride around the lake. Customers can be either student, staff, or members of the public. Some of the staffs also can be a student at a same time. Customers must provide their details (such as IC number, name, address and phone number) to the counter and the counter will generate the booking information for the customers and provide the amount of payment based on duration used by them. A special discount is given to the students. Staff can only pay the rent via salary deduction which will be accumulated at the end of the month."

a) Draw the enhanced entity relational diagram (EERD) based on the given scenario.

(6 Marks)

b) Chicken Black Pepper and Fish Curry are the top two among various menu included in Pak Lah's Café. Since the café allowed the customer to customize the menu, sometime the customer love to switch the menu into Fish Black Pepper and Chicken Curry. Based on the specialization and generalization concept, complete the **Diagram 1** below

(4 Marks)

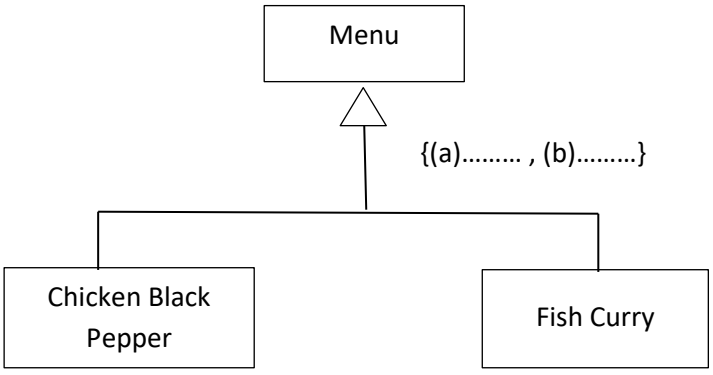


Diagram 1

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QUESTION 2 (20 MARKS)

Answer the questions (a), (b), (c) and (d) using SQL statements based on the following tables, which consist of THREE (3) relations: **Student**, **Subject** and **Lecturer**.

Student (SID, SName, SGender, SDob, SAddress, STelNo)

Table 2.1: Student relation

SID	SName	SGender	SDob	SAddress	STelNo
111	Ali	Male	01-01-2001	Tmn. Universiti, Skudai, Johor	075211234
222	Ani	Female	12-10-1998	Tmn. Pulai Perdana, Johor	075217894
333	Wahab	Male	03-08-2000	Tmn. Pulai Jaya, Johor	
444	Yan	Male	08-06-1999	Tmn. Pulai Emas, Johor	

Subject (SID, SubName, LID)

Table 2.2: Subject relation

SID	SubName	LID
111	Database	11
333	Database	11
111	Data Structure	33
444	Data Structure	22

Lecturer (LID, LName, LGender, LDob, LAddress, LTelNo)

Table 2.3: Lecturer relation

LID	LName	LGender	LDob	LAddress	LTelNo
L11	Aminah	Female	01-01-1959	Tmn. Universiti, Skudai, Johor	075214444
L22	Jenny	Female	12-16-1968	Tmn. Pulai Perdana, Johor	075217878
L33	Hamid	Male	03-10-1957	Tmn. Pulai Jaya, Johor	075214563
L34	Amni	Female	03-11-1982	Tmn Pulai Indah, Johor	075548888

Based on the relations above, answer ALL questions below.

a) Write the SQL statement to add a row of data below into Student relation. (2 Marks)

SID	SName	SGender	SDob	SAddress	STelNo
555	Abu	Male	05-12-2001	Tmn. Pulai Flora, Johor	-

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b) Write the SQL statement to modify student telephone number existing values for a student with ID 333 to 012-3456789 in a Student relation. (2 Marks)

c) Write the SQL statement to list student information (id number and name) with the subject information (subject name and lecturer id). (3 Marks)

d) Write the SQL statement to retrieve all columns and rows from Lecturer relation that meet both condition including lecturer gender is female and the third letter of the lecturer name is the letter 'n'. (3 Marks)

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Table 2.5 shows the instances of a relation named Employee. Answer the questions (e), (f), (g) and (h) based on **Table 2.5.**

Table 2.5: Employee relation

EmployeeNo	fName	lName	position	salary	branchNo
SL70	Ahmad	Lim	Assistant	9000.00	AA007
SB18	Mary	Ann	Manager	NULL	AA003
SB48	Intan	Arsyila	Assistant	5000.00	AA010
SG11	Norliza	Sahari	Supervisor	17000.00	AA007
SL59	Zizan	Nabil	Assistant	6500.00	AA010
SG37	James	Ian	Manager	24000.00	AA007

e) Show the output for the following query: (2 Marks)

```
SELECT DISTINCT EmployeeNo, fName, lName
FROM Employee
WHERE salary < 15000 ORDER BY fName DESC;
```

f) Show the result of the following SQL statement: (2 Marks)

```
SELECT COUNT (salary)
FROM Employee;
```

g) Show the output of the following SQL statement: (3 Marks)

```
SELECT MIN(salary) AS salaryMIN, MAX(salary) AS
salaryMAX, AVG(salary) AS salaryAVG
FROM Employee;
```

h) Produce the result of the following SQL statement: (3 Marks)

```
SELECT branchNo, COUNT(EmployeeNo) AS myCount,
SUM(salary) AS salarySUM
FROM Employee
GROUP BY branchNo
HAVING COUNT(EmployeeNo) > 1
ORDER BY branchNo;
```

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QUESTION 3 (20 MARKS)

Answer the following questions based on the conceptual ERD of a project management system as shown in **Figure 3.1**.

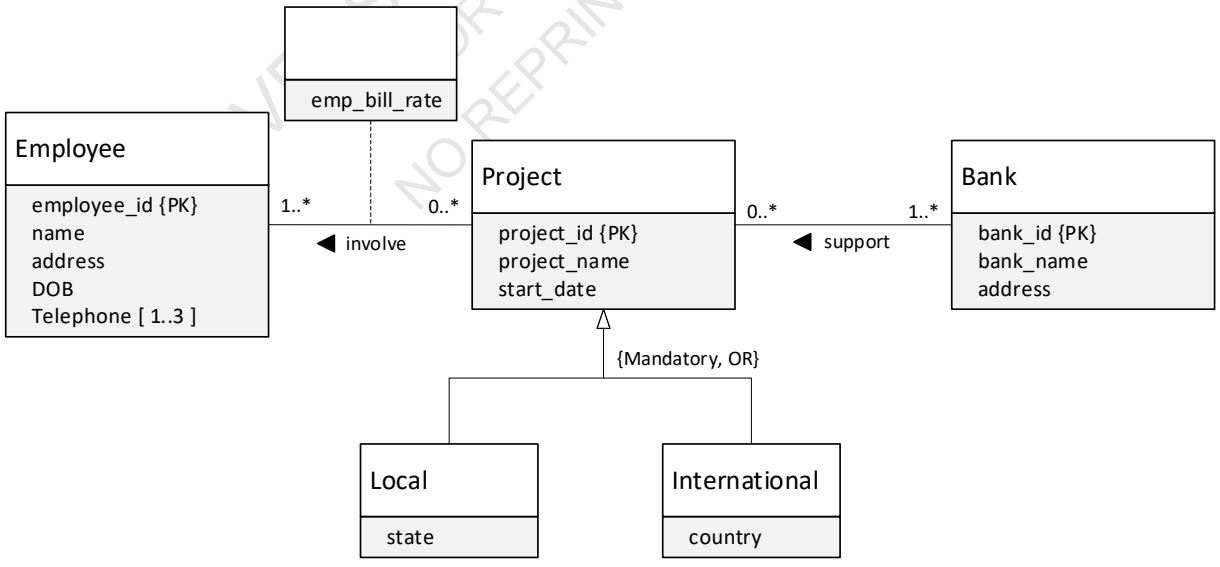


Figure 3.1: Conceptual ERD of a project management system.

a) Draw the logical ERD from the conceptual ERD shown in **Figure 3.1.** (10 Marks)

b) Based on the logical ERD in (a), derive the relational schemas. (10 Marks)

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QUESTION 4 (15 MARKS)

Table 4.1 lists the loan application data. Each application is handled by a specific banker. A client can apply for multiple types of loan from the bank.

Table 4.1: Loan application data

clientNo	clientName	staffNo	staffName	loanType	loanAmount	applicationDate
C0011	Hilmi	S0021	Ahmad	Car Loan	RM80000.00	12 Oct 2018
C0012	James	S0021	Ahmad	Personal Loan Personal Loan	RM20000.00 RM15000.00	30 Sep 2018 28 Sep 2018
C0013	Jones	S0021	Ahmad	House Loan	RM600000.00	12 Oct 2018
C0011	Hilmi	S0033	Sue Ann	House Loan	RM560000.00	17 Aug 2018
C0014	Amirah	S0033	Sue Ann	Car Loan	RM76000.00	16 Sep 2018
C0013	Jones	S0042	Joseph	Car Loan	RM60000.00	15 Oct 2018
C0011	Hilmi	S0042	Joseph	Personal Loan	RM30000.00	29 Aug 2018

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Based on the table above, answer the following questions:

a) Identify **ONE** anomaly that exists in the table 4.1 above and provide the example based on the data in table above. (2 marks)

b) Identify the level of Normal Form for Table 4.1. Justify your answer. (2 marks)

c) Identify all the functional dependencies that can be derived from Table 4.1. (4 marks)

d) Normalize the relation in Table 4.1 up to Third Normal Form (3NF). Illustrate the process of the normalization and state any assumptions.

(7 marks)

PART B : CASE STUDY QUESTIONS [35 MARKS]

INSTRUCTION : ANSWER ALL QUESTIONS.
Please write your answer in the answer booklet.

The Aafiyat chain of pharmacies has offered to give you a free lifetime supply of medicine if you design its database. Given the rising cost of health care, you agree. Here is the information that you gather:

- Patients are identified by a patient ID, and their names, addresses, and ages must be recorded.
- Doctors are identified by a dr_ID. For each doctor, the name, specialty, and years of experience must be recorded.
- Each pharma drug supplier is identified by name and has maximum three phone number.
- For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharma drug supplier, and the trade name identifies a drug uniquely from among the products of that company. If a pharma drug supplier is deleted, you need not keep track of its products any longer.
- Each pharmacy has a pharmacy ID, name, address, and phone number.
- Every patient has a primary physician. Every doctor has at least one patient.
- Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.
- Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it. You can assume that, if a doctor prescribes the same drug for the same patient more than once, only the last such prescription needs to be stored.
- Pharma drug supplier have long-term contracts with pharmacies. A pharma drug supplier can contract with several pharmacies, and a pharmacy can contract with several pharma drug supplier. For each contract, you have to store a start date, an end date, and the text of the contract.

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- Pharmacies appoint a supervisor for each contract. There must always be a supervisor for each contract, but the contract supervisor can change over the lifetime of the contract.

(a) Identify all entities and attributes for each entity of this case study. Identify the Primary Keys. Provide your answer in the following format.

ENTITY	ATTRIBUTE

Table B1: Aafiyat Pharmacy entity and attributes

(10 marks)

(b) Using your answers for (a) and (b), draw a single ER diagram to represent the data requirements of Aafiyat Pharmacy. Determine the multiplicity constraints for each relationship described in (b). Represent the multiplicity for each relationship in the ER diagram. State any assumptions necessary to support your design.

(15 marks)

(c) Using your answer for (b), derive the relations.

(10 marks)

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